

Sydney Technical High School



Extension One Mathematics HSC Assessment Task 2 March 2011

Name.....

Teacher.....

General Instructions

- Working Time – 70 minutes.
- Write using a blue or black pen.
- Approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a new page.

Total marks (60)

- Attempt Questions 1-6.
- All questions are of equal value.
- Mark values are shown with the questions

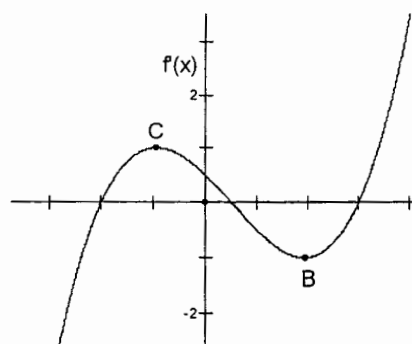
Question	1	2	3	4	5	6	TOTAL
Mark							

Question 1 (10 marks)**Marks**

- a) Find the primitive function of $\frac{3}{4\sqrt{x}}$ 1
- b) Consider the curve $y = x^4 - \frac{4}{3}x^3 - 2x^2 + 4x + 3$
- (i) Obtain y' and y'' for this function 2
- (ii) Find the stationary points. 2
- (iii) Determine the nature of each of the stationary points. 2
- (iv) Find the x coordinates of the two points of inflexion. 1
- (v) Sketch the curve for the domain 2

Question 2 (10 marks) Begin a SEPARATE sheet of paper

- a) The graph of $y = f'(x)$ is shown. The zeros of $f'(x)$ are $x = -2, 0.5$, and 3
C has x coordinate -1 and B has x coordinate 2



- (i) For what values of x is $f(x)$ increasing? 1
- (ii) C is a local maximum on $f'(x)$.
What type of point occurs on $f(x)$ at the same x value as that shown at C.
Justify your answer. 2
- (iii) For what values of x is $f(x)$ concave down? 1
- c) $g'(x) = 3x^2 - 4 + \frac{1}{x^2}$ 3
 $g(x)$ takes the value 4 when $x = 1$. Find $g(x)$.
- d) Evaluate $\int_1^2 \left(x^2 + \frac{1}{x^3} \right) dx$ 3

Question 3 (10 marks) Begin a SEPARATE sheet of paper**Marks**

- a) $y = f(x)$ is a continuous function and has a table of values as shown below.

4

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
$f(x)$	2.3	2.5	3.1	2.7	2.4	2.1	1.6

Use the Trapezoidal rule to find the approximate value of $\int_1^4 f(x) dx$ correct to one decimal place.

- b) Two sailors are paid to bring a motor launch back to Sydney from Gilligans Island, a distance of 1 200 km. They are each paid \$25 per hour for the time spent at sea.

The launch uses fuel at a rate $R = 20 + \frac{v^2}{10}$ litres per hour. Diesel costs \$1.25 per L and (v) is the velocity in km/hour.

- (i) Show that, to bring the launch back from Gilligans Island,

3

the total cost to the owners is $\frac{90000}{v} + 150v$.

- (ii) Find the speed which minimises the cost and determine this cost.

3**Question 4** (10 marks) Begin a SEPARATE sheet of paper

- a) Use Simpson's rule with 5 function values to evaluate

3

$$\int_0^4 \frac{\sqrt{144 - 9x^2}}{4} dx$$

- b) Consider the functions $y = 3 - \frac{x}{2}$ and $y = \frac{1}{2}x^2 - 2x + 1$

- (i) Find the x values where the curves intersect.

2

- (ii) Find the area between the curves.

2

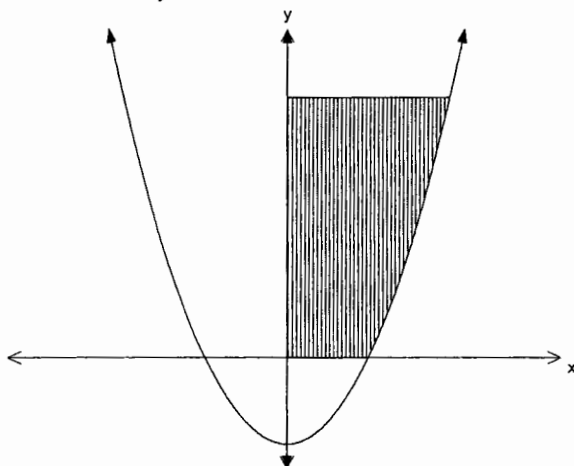
- c) Using the substitution $u = 2x^2 - 3x$, or otherwise, find $\int \frac{(4x-3)dx}{\sqrt{2x^2-3x}}$

3

Question 5 (10 marks) Begin a SEPARATE sheet of paper

Marks

- a) The diagram shows the region bounded by the curve $y = 2x^2 - 2$, the line $y = 6$ and the x and y axes. **4**

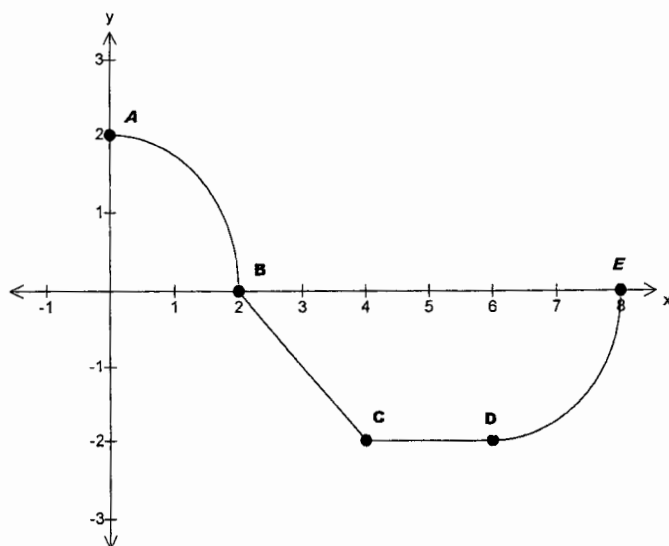


Find the volume of the solid of revolution formed when the region is rotated about the y axis.

- b) Evaluate $\int_3^{18} \frac{x}{\sqrt{x-2}} dx$ using a suitable substitution. **3**

- c) The region, enclosed by the parabola $y^2 = 4ax$ and the line $x = a$, is rotated about the x -axis. Find the volume of the solid formed. **3**

Question 6 (10 marks) Begin a SEPARATE sheet of paper

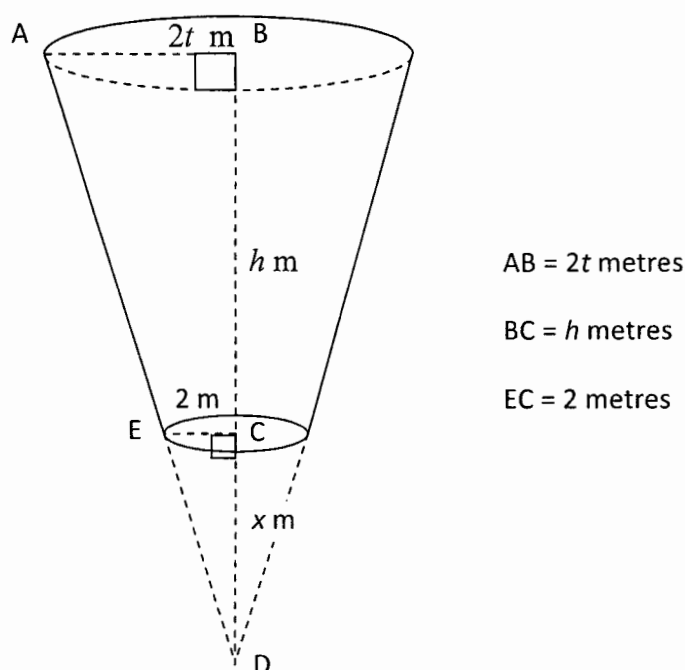


- a) The graph of the function f consists of a quarter circle AB, a straight line segment BC, a horizontal straight line segment CD, and a quarter circle DE as shown above.

(i) Evaluate $\int_0^8 f(x) dx$ 2

- (ii) For what values of x satisfying $0 < x < 8$ is the function f NOT differentiable 1

- b) A truncated cone is to be used as a part of a hopper for a grain harvester. It has a total height of h metres. The top radius is to be t times greater than the bottom radius which is 2 metres.



- i) If x is the height of the removed section of the original cone, show using similar triangles that $x = \frac{h}{t-1}$ 2
- ii) Show that the volume of the truncated cone is given by $V = \left(\frac{4\pi h}{3}\right)(t^2 + t + 1)$ 2
- iii) If the upper radius plus the lower radius plus the height of the truncated cone must total 12 metres, calculate the maximum volume of the hopper. 3

END OF EXAMINATION



STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$i) \int \frac{3}{4\sqrt{x}} dx = \int \frac{3}{4} x^{-\frac{1}{2}} dx$$

$$= \frac{3\sqrt{x}}{2} + c \quad (1)$$

$$ii) y = x^4 - \frac{4}{3}x^3 - 2x^2 + 4x + 3 \quad (1)$$

$$y' = 4x^3 - 4x^2 - 4x + 4 \quad (1)$$

$$y'' = 12x^2 - 8x - 4 \quad (1)$$

$$iii) 4x^3 - 4x^2 - 4x + 4 = 0$$

$$x^2(x-1) - 1(x-1) = 0$$

$$(x-1)(x+1)(x-1) = 0$$

$$\left\{ \begin{array}{l} x = 1 \quad x = -1 \\ y = 4\frac{2}{3} \quad y = -\frac{2}{3} \end{array} \right\} \quad (1)$$

$$iii) \text{ When } x = -1, y'' > 0$$

$$\therefore \text{ minimum at } (-1, -\frac{2}{3}) \quad (1)$$

$$\text{ When } x = 1, y'' = 0$$

$$\therefore \text{ horizontal point of inflexion at } (1, 4\frac{2}{3}) \quad (1)$$

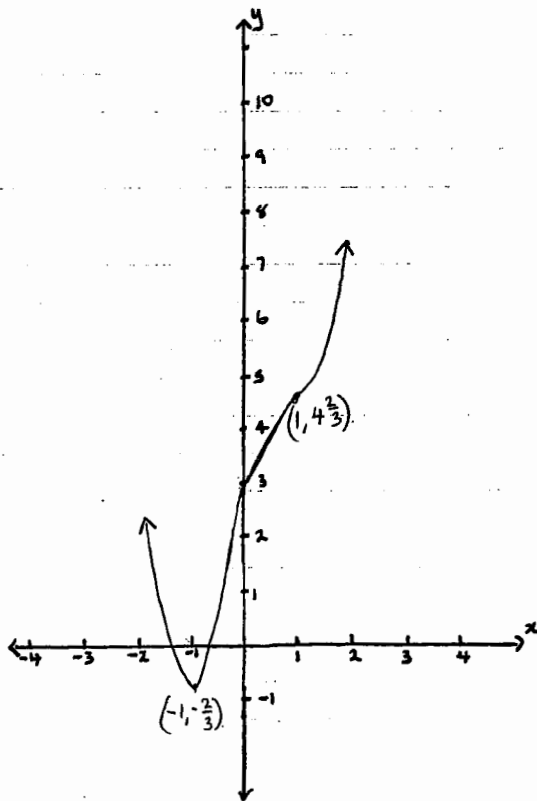
$$v) \text{ Points of inflexion occur when } y'' = 0$$

$$12x^2 - 8x - 4 = 0$$

$$3x^2 - 2x - 1 = 0$$

$$(3x+1)(x-1) = 0$$

$$x = -\frac{1}{3} \text{ or } 1 \quad (1)$$



Question 2

$$ai) f(x) \text{ is increasing where } f'(x) > 0$$

$$\text{ie } -2 < x < \frac{1}{2} \text{ and } x > 3 \quad (1)$$

$$ii) \text{ A point of inflexion, since } c \text{ has max. gradient between } x = -2 \text{ and } x = 0.5 \text{ which are stat points } (f'(x) = 0) \quad (1)$$

$$iii) f(x) \text{ will be concave down when } f'(x) \text{ is decreasing}$$

$$-1 < x < 2 \quad (1)$$

$$b. g(x) = \int g'(x) dx$$

$$= x^3 - 4x - x^{-1} + c \quad (1)$$

$$g(1) = 4$$

$$4 = 1^3 - 4 \times 1 - 1^{-1} + c$$

$$c = 8$$

$$g(x) = x^3 - 4x - \frac{1}{x} + 8 \quad (1)$$

$$c. \int_1^2 \left(x^2 + \frac{1}{x^3} \right) dx$$

$$= \left[\frac{x^3}{3} - \frac{1}{2x^2} \right]_1^2$$

$$= \left(\frac{8}{3} - \frac{1}{8} \right) - \left(\frac{1}{3} - \frac{1}{2} \right)$$

$$= \frac{65}{24}$$

$$= 2\frac{17}{24}$$

Question 3

$$a) \int_1^4 f(x) dx \approx \frac{h}{2} [y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4) + 4y_5]$$

$$\approx \frac{0.5}{2} [2.3 + 1.6 + 2(2.5 + 3.1 + 2.7 + 2.4 + 2.1) + 4 \times 0.5]$$

$$\approx \frac{1}{4} [3.9 + 2(12.8)] \quad (1)$$

$$\approx \frac{1}{4} [3.9 + 25.6] \quad (1)$$

$$\approx 7.4 \text{ unit}^2 \text{ (1 dp)} \quad (1)$$

$$bi) \text{ Time to complete the trip } \frac{1200}{v} \text{ and sailors paid } \$50/h$$

$$\text{Cost} = \left[20 + \frac{v^2}{10} \right] \times \frac{1200}{v} \times 1.25 + 50 \times \frac{1200}{v}$$

$$\text{Cost} = \frac{1200}{v} \left[75 + \frac{1.25v^2}{10} \right]$$

$$\text{Cost} = \frac{90000}{v} + 150v \quad (1)$$

$$bii) \frac{d(\text{Cost})}{dv} = 150 - \frac{90000}{v^2} = 0$$

$$\text{When } v^2 = 600$$

$$v = 24.495 \text{ km/h} \quad (1)$$

$$\frac{d^2(\text{Cost})}{dv^2} = 180000v^{-3} \text{ at } v = 24.495$$

$$\frac{180000}{24.495^3} > 0$$

$$\therefore \text{ min} \quad (1)$$

$$\therefore \text{ Cost} = \frac{90000}{24.495} + 150 \times 24.495$$

$$= \$7348.47$$

Question 4

$$\approx \frac{h}{3} [f(0) + f(4) + 2xf(2) + 4(f(1) + f(3))] \\ \approx \frac{1}{3} \left[3 + 0 + \frac{2x\sqrt{108}}{4} + \frac{4x\sqrt{135}}{4} + \frac{\sqrt{63}}{4} \right] \\ \approx 9.2507855$$

i) $y = 3 - \frac{x}{2}$
 $y = \frac{1}{2}x^2 - 2x + 1$

$$3 - \frac{x}{2} = \frac{x^2}{2} - 2x + 1 \quad (1)$$

$$6 - x = x^2 - 4x + 2$$

$$0 = x^2 - 3x - 4$$

$$0 = (x-4)(x+1)$$

$$x = -1 \quad x = 4$$

ii) $\int_{-1}^4 3 - \frac{x}{2} - \left(\frac{x^2}{2} - 2x + 1 \right) dx$

$$= \int_{-1}^4 2 + \frac{3x}{2} - \frac{x^2}{2} dx$$

$$= \left[2x + \frac{3x^2}{4} - \frac{x^3}{6} \right]_{-1}^4$$

$$= \left(8 + 12 - \frac{32}{3} \right) - \left(\frac{3}{4} + \frac{1}{6} - 2 \right)$$

$$= 10\frac{5}{12}$$

4c) $\int \frac{4x-3}{\sqrt{2x^2-3x}} dx$

$$u = 2x^2 - 3x$$

$$\frac{du}{dx} = 4x - 3 \quad (1)$$

$$\therefore \int \frac{4x-3}{\sqrt{2x^2-3x}} = \int \frac{du}{\sqrt{u}} = \int u^{-\frac{1}{2}} du$$

$$= 2u^{\frac{1}{2}} + c$$

$$= 2(2x^2 - 3x)^{\frac{1}{2}} + c$$

$$= 2\sqrt{2x^2 - 3x} + c$$

Question 5

1a) $y = 2x^2 - 2$
 $v = \pi \int_0^6 x^2 dy$

$$= \pi \int_0^6 \frac{y+2}{2} dy$$

$$= \pi \left[\frac{y^2}{4} + y \right]_0^6 \quad (1)$$

$$= \pi \left[\left(\frac{36}{4} + 6 \right) - 0 \right]$$

$$= 15\pi \text{ units}^3 \quad (1)$$

b) $\int_3^{18} \frac{x}{\sqrt{x-2}} dx$ let $u = \sqrt{x-2}$
 $u^2 = x-2$

$$2u \frac{du}{dx} = 1$$

$$dx = 2u du$$

$$\therefore x = u^2 + 2$$

When $x = 3$ $u = 1$
 $x = 18$ $u = 4$

$$= 2 \int_1^4 \frac{(u^2+2)u du}{u}$$

$$= 2 \int_1^4 (u^2 + 2) du \quad \left| = 2 \left[\frac{u^3}{3} + 2u \right]_1^4 \right.$$

$$= 54$$

Question 5 continued

5c) $v = \pi \int_0^a y^2 dx$ (1)

$$v = \pi \int_0^a 4ax dx$$

$$v = \pi \left[2ax^2 \right]_0^a \quad (1)$$

$$v = \pi [2a^3 - 0]$$

$$v = 2\pi a^3 \text{ units}^3 \quad (1)$$

Question 6

ai) $\int_0^8 f(x) dx = -\left(\frac{1}{2} \times 2 \times 2\right) - 2 \times 2$
 $= -6$ (2)

aii) The function is NOT differentiable at $x=2$ and $x=4$
 (the end points are NOT included at $x=6$, the gradient is continuous) (1)

b i) In $\triangle ABD$ and $\triangle ECD$

$$\frac{2t}{h+2} = \frac{2}{x}$$

$$2tx = 2(h+x) \quad (1)$$

$$2tx = 2h + 2x$$

$$2tx - 2x = 2h$$

$$2x(t-1) = 2h$$

$$x = \frac{h}{t-1} \quad (1)$$

b ii) $V = \frac{1}{3} \pi (2t)^2 \cdot (h+x) - \frac{1}{3} \pi 2^2 x$

$$= \frac{1}{3} \pi (2t)^2 \left(h + \frac{h}{t-1} \right) - \frac{1}{3} \pi (2)^2 \left(\frac{h}{t-1} \right) \quad (1)$$

$$= \frac{1}{3} \pi (2t)^2 \left(\frac{ht}{t-1} \right) - \frac{1}{3} \pi (2)^2 \left(\frac{h}{t-1} \right)$$

$$= \frac{1}{3} \pi (2)^2 \left(\frac{h}{t-1} \right) (t^3 - 1)$$

$$= \frac{4}{3} \pi \left(\frac{h}{t-1} \right) (t-1) (t^2 + t + 1)$$

$$= 4\pi h (t^2 + t + 1) \quad (1)$$

ii) Sum of radii and height = 12
 $2 + h + 2t = 12$
 $h = 10 - 2t$ (1)

$$v = \frac{4\pi h}{3} (t^2 + t + 1)$$

$$v = \frac{4\pi}{3} (10 - 2t) (t^2 + t + 1) \quad (1)$$

$$v = \frac{4\pi}{3} (10t^2 + 10t + 10 - 2t^3 - 2t^2 - 2t)$$

$$v = \frac{4\pi}{3} (8t^2 + 8t - 2t^3 + 10) \quad (1)$$

$$\frac{dv}{dt} = \frac{4\pi}{3} (16t + 8 - 6t^2) = 0$$

$$16t + 8 - 6t^2 = 0$$

$$t = \frac{-16 \pm \sqrt{16^2 - 4 \times 6 \times 8}}{2 \times -6}$$

$$t = \frac{-16 \pm \sqrt{448}}{-12}$$

$$t = -0.43 \text{ or } 3.10$$

$$\frac{d^2v}{dt^2} = \frac{4\pi}{3} (16 - 12(3.10))$$

$$= -88.7$$

$$\therefore \frac{d^2v}{dt^2} < 0$$

$\therefore V$ is a maximum

$$v = \frac{4\pi}{3} [8 \times 3.10^2 + 8 \times 3.10 - 2 \times 3.10^3 + 10]$$

$$= 218.2$$